Task-20:- Technical Documentation(Self-hosted DNS server and an AWS Route 53 hosted zone)

1. Introduction

1.1 Purpose

This document describes the implementation of two DNS systems:

- Self-hosted DNS server using BIND9 on AWS EC2
- Managed DNS service using AWS Route 53

It demonstrates how DNS resolution changes when Nameserver (NS) records are updated.

1.2 Scope

Included:

- EC2 DNS server setup
- Route 53 hosted zone configuration
- DNS record creation
- NS switching
- DNS testing

Not Included:

- Domain registration
- DNSSEC
- Advanced routing policies

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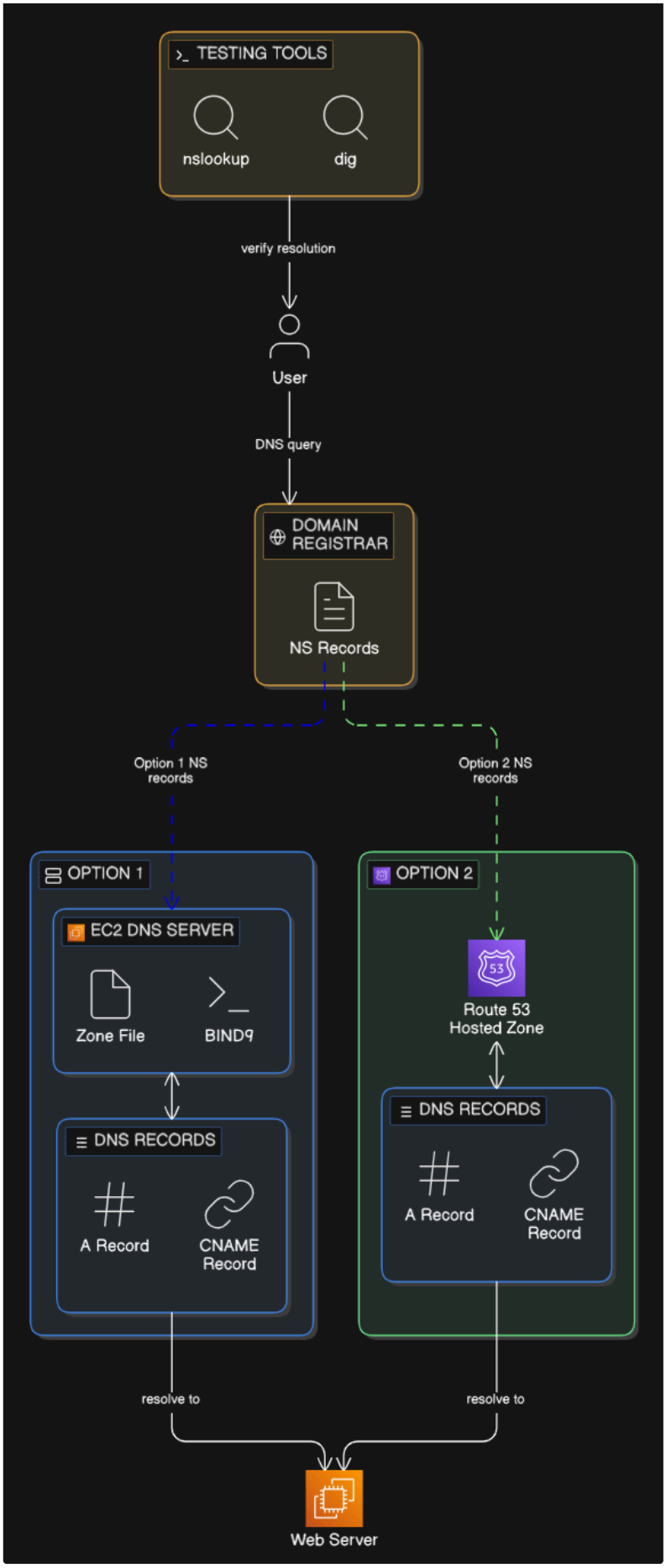
2. Architecture Overview


TESTING TOOLS
nslookup
dig
verify resolution
User
DNS query
DOMAIN REGISTRAR
NS Records
Option 1 NS records
Option 2 NS records
OPTION 1
EC2 DNS SERVER
Zone File
BIND9
DNS RECORDS
A Record
CNAME Record
OPTION 2
Route 53 Hosted Zone
DNS RECORDS
A Record
CNAME Record
resolve to
resolve to
Web Server

DNS Flow Explanation

1. User sends DNS query
2. Query goes to domain registrar
3. Registrar forwards to configured NS
4. Selected DNS server responds

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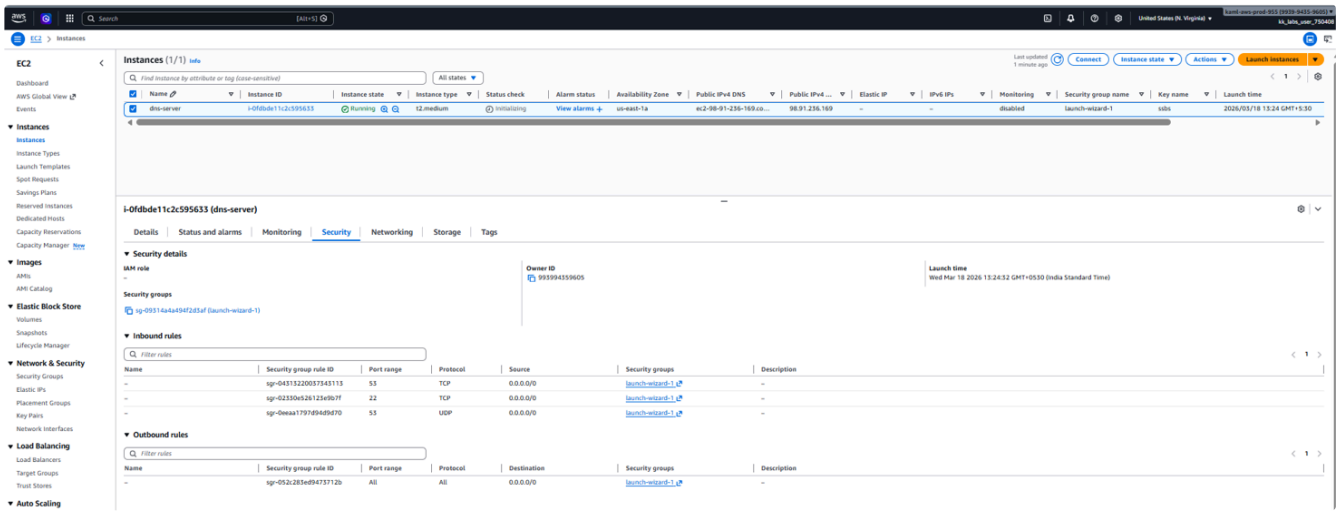
3. Implementation Steps

🔹 3.1 Self-Hosted DNS Setup (BIND9 on EC2)

Step 1: Launch EC2 Instance

- Created Ubuntu EC2 instance
- Allowed ports:
  - SSH (22)
  - DNS (53 TCP/UDP)



Step 2: Install BIND9

```
sudo apt update
sudo apt install bind9 -y
```

```
ubuntu@ip-172-31-41-127:~$ sudo apt update
sudo apt install bind9 -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1837 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [336 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1566 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [319 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2819 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [652 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1540 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7044 B]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [496 B]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7348 B]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.1 kB]
Get:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:38 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:39 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:40 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:41 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:42 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:43 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [246 kB]
Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:45 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.5 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [219 kB]
```

Step 3: Configure Zone

Edited:

```
/etc/bind/named.conf.local
```

Added zone configuration for domain.

```
ubuntu@ip-172-31-41-127: ~
  GNU nano 7.2                                        /etc/bind/named.conf.local *
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
zone "vraj.pushkar.cloud" {
    type master;
    file "/etc/bind/db.vraj.pushkar.cloud";
};
```

Step 4: Create Zone File

Created:

```
/etc/bind/db.vraj.pushkar.cloud
```

Configured records:

- A record
- NS record
- CNAME record

```
  GNU nano 7.2                                    /etc/bind/db.vraj.pushkar.cloud *
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA      ns1.vraj.pushkar.cloud. admin.vraj.pushkar.cloud. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
; Name Server
@       IN      NS      ns1.vraj.pushkar.cloud.

; NS A Record
ns1     IN      A       98.91.236.169

; Main Domain
@       IN      A       98.91.236.169

; Subdomain
www     IN      A       98.91.236.169

; CNAME Example
blog    IN      CNAME   www
```

Step 5: Restart DNS Server

```
sudo systemctl restart bind9
```

```
ubuntu@ip-172-31-41-127:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-18 07:56:45 UTC; 28s ago
       Docs: man:named(8)
   Main PID: 1815 (named)
     Status: "running"
      Tasks: 8 (limit: 4663)
     Memory: 24.0M (peak: 24.4M)
        CPU: 61ms
     CGroup: /system.slice/named.service
             └─1815 /usr/sbin/named -f -u bind

Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './DNSKEY/IN': 2001:7fd::1#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './NS/IN': 2001:7fd::1#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10::b#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './NS/IN': 2801:1b8:10::b#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './DNSKEY/IN': 2001:500:9f::42#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: managed-keys-zone: Initializing automatic trust anchor management for zone '.'; DNSKEY>
Mar 18 07:56:45 ip-172-31-41-127 named[1815]: managed-keys-zone: Initializing automatic trust anchor management for zone '.'; DNSKEY>

ubuntu@ip-172-31-41-127:~$ sudo nano /etc/bind/named.conf.local
ubuntu@ip-172-31-41-127:~$ sudo cp /etc/bind/db.local /etc/bind/db.vraj.pushkar.cloud
sudo nano /etc/bind/db.vraj.pushkar.cloud
ubuntu@ip-172-31-41-127:~$ sudo systemctl restart bind9
```

Step 6: Test Self-Hosted DNS

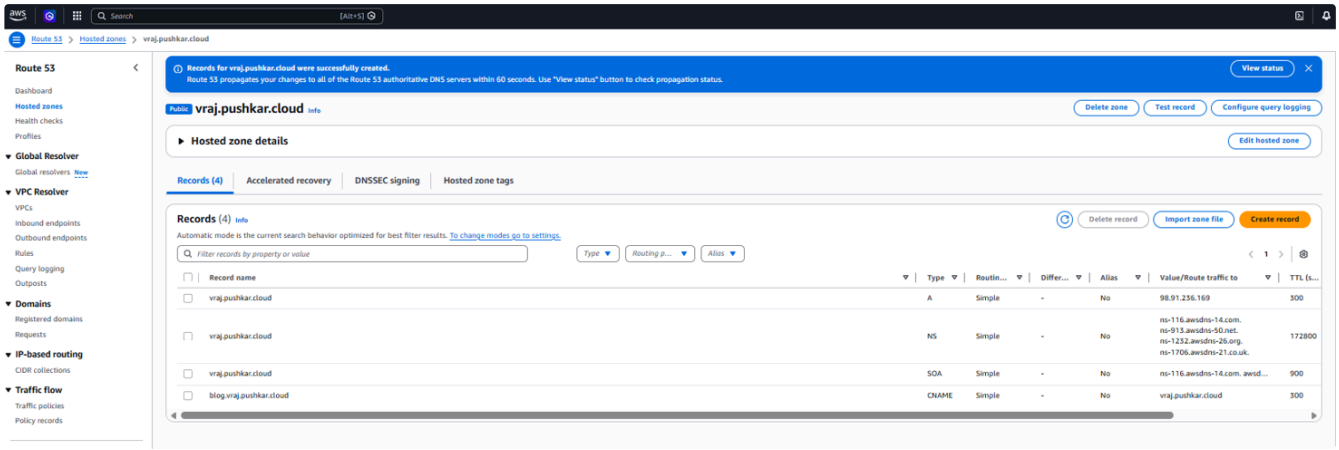
```
dig @EC2_IP vraj.pushkar.cloud
```

```
ubuntu@ip-172-31-41-127:~$ dig @98.91.236.169 vraj.pushkar.cloud

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @98.91.236.169 vraj.pushkar.cloud
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 40562
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 77defbf3f3b2f8e70100000069ba621eb48423f7ebb6b7e2 (good)
;; QUESTION SECTION:
;vraj.pushkar.cloud.            IN      A

;; ANSWER SECTION:
vraj.pushkar.cloud.     604800  IN      A       98.91.236.169

;; Query time: 0 msec
;; SERVER: 98.91.236.169#53(98.91.236.169) (UDP)
;; WHEN: Wed Mar 18 08:28:14 UTC 2026
;; MSG SIZE  rcvd: 91
```

🔹 3.2 AWS Route 53 Setup

Step 1: Create Hosted Zone

- Opened Route 53
- Created public hosted zone



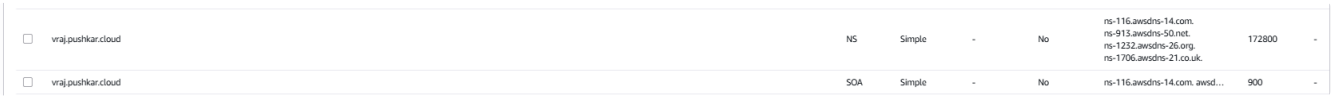
Step 2: Add DNS Records

Created:

- A record → EC2 IP
- CNAME → blog subdomain

Step 3: Note NS Records

Copied AWS nameservers for domain mapping.



🔹 3.3 Nameserver (NS) Configuration

Step 1: Point to Self-Hosted DNS

- Updated NS at domain registrar
- Pointed to EC2 DNS server

Step 2: Test DNS Resolution

```
dig vraj.pushkar.cloud
```

```
ubuntu@ip-172-31-41-127:~$ dig vraj.pushkar.cloud

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> vraj.pushkar.cloud
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31817
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;vraj.pushkar.cloud.            IN      A

;; ANSWER SECTION:
vraj.pushkar.cloud.     261     IN      A       98.91.236.169

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Mar 18 08:24:27 UTC 2026
;; MSG SIZE  rcvd: 63
```

🔹 3.4 DNS Testing and Verification

Test 1: Direct Query to Route 53

```
dig @ns-116.awsdns-14.com vraj.pushkar.cloud
```

```
ubuntu@ip-172-31-41-127:~$ dig @ns-116.awsdns-14.com vraj.pushkar.cloud

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @ns-116.awsdns-14.com vraj.pushkar.cloud
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25352
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 4, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;vraj.pushkar.cloud.            IN      A

;; ANSWER SECTION:
vraj.pushkar.cloud.     300     IN      A       98.91.236.169

;; AUTHORITY SECTION:
vraj.pushkar.cloud.     172800  IN      NS      ns-116.awsdns-14.com.
vraj.pushkar.cloud.     172800  IN      NS      ns-1232.awsdns-26.org.
vraj.pushkar.cloud.     172800  IN      NS      ns-1706.awsdns-21.co.uk.
vraj.pushkar.cloud.     172800  IN      NS      ns-913.awsdns-50.net.

;; Query time: 6 msec
;; SERVER: 205.251.192.116#53(ns-116.awsdns-14.com) (UDP)
;; WHEN: Wed Mar 18 08:27:25 UTC 2026
;; MSG SIZE  rcvd: 203
```

Test 2: Direct Query to Self-Hosted DNS

```
dig @EC2_IP vraj.pushkar.cloud
```

Test 3: DNS Trace

```
dig vraj.pushkar.cloud +trace
```

```
ubuntu@ip-172-31-41-127:~$ dig vraj.pushkar.cloud +trace

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> vraj.pushkar.cloud +trace
;; global options: +cmd
.                       300     IN      NS      m.root-servers.net.
.                       300     IN      NS      a.root-servers.net.
.                       300     IN      NS      b.root-servers.net.
.                       300     IN      NS      c.root-servers.net.
.                       300     IN      NS      d.root-servers.net.
.                       300     IN      NS      e.root-servers.net.
.                       300     IN      NS      f.root-servers.net.
.                       300     IN      NS      g.root-servers.net.
.                       300     IN      NS      h.root-servers.net.
.                       300     IN      NS      i.root-servers.net.
.                       300     IN      NS      j.root-servers.net.
.                       300     IN      NS      k.root-servers.net.
.                       300     IN      NS      l.root-servers.net.
;; Received 239 bytes from 127.0.0.53#53(127.0.0.53) in 2 ms

;; UDP setup with 2001:500:2d::d#53(2001:500:2d::d) for vraj.pushkar.cloud failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:500:2d::d#53(2001:500:2d::d) for vraj.pushkar.cloud failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:500:2d::d#53(2001:500:2d::d) for vraj.pushkar.cloud failed: network unreachable.
;; UDP setup with 2001:76d::1#53(2001:76d::1) for vraj.pushkar.cloud failed: network unreachable.
;; UDP setup with 2001:500:9f::42#53(2001:500:9f::42) for vraj.pushkar.cloud failed: network unreachable.
cloud.                  172800  IN      NS      ns01.trs-dns.com.
cloud.                  172800  IN      NS      ns01.trs-dns.net.
cloud.                  172800  IN      NS      ns10.trs-dns.org.
cloud.                  172800  IN      NS      ns10.trs-dns.info.
cloud.                  86400   IN      DS      7041 13 2 03E1B41319FDD3A14E27CE35FFBA8036DA51E328165CE102AE69A6C1 203F64E4
cloud.                  86400   IN      RRSIG   DS 8 1 86400 20260331050000 20260318040000 21831 . DHXiv3qRJIPZclkN+rJGF4YG/8awQoB5i5oFqxG5Bj7rWauMzgHqHxFF fzAAobm7kkKOyirmm+NRldRYNDbpPpqirL9z4VNplIG+dKooL/axcXtM Z3xA8y+pHY2jUDQxTQzVQ
mZJjpuE4aJpvz0Ib3z0khlSiwG1ARiU2b7e nvvbaNI70RO1Za5McdllEys4c0p0d3XW2IWk+ccc8MRCl+v4RLjJeLT7 zhJ1ntCCqJjrJMK4jxhLW//3KVgzafGIWCaNgT7oDGHLcycANAKxsBb2 krAFn3yYPBOu1e+9zpy8Bm8mZJ55GcnVaM0d/SDyKxYbzjYngJcz79NQ g6kgkQ==
;; Received 679 bytes from 192.58.128.30#53(j.root-servers.net) in 1 ms

;; UDP setup with 2620:171:813:1534:8::1#53(2620:171:813:1534:8::1) for vraj.pushkar.cloud failed: network unreachable.
;; UDP setup with 2620:57:4002::1#53(2620:57:4002::1) for vraj.pushkar.cloud failed: network unreachable.
;; UDP setup with 2620:171:812:1534:8::1#53(2620:171:812:1534:8::1) for vraj.pushkar.cloud failed: network unreachable.
pushkar.cloud.          900     IN      NS      ns-1234.awsdns-26.org.
pushkar.cloud.          900     IN      NS      ns-1637.awsdns-12.co.uk.
pushkar.cloud.          900     IN      NS      ns-482.awsdns-60.com.
pushkar.cloud.          900     IN      NS      ns-984.awsdns-59.net.
HT3BBE52QFEA3EKHEL6VAE4CVUE8777V.cloud. 300 IN NSEC3 1 1 0 - HT4FGH4PBH0ALU4PC8DK05VSHJ8PKLAF NS SOA RRSIG DNSKEY NSEC3PARAM
HT3BBE52QFEA3EKHEL6VAE4CVUE8777V.cloud. 300 IN RRSIG NSEC3 13 2 300 20260407013906 20260309141535 36804 cloud. Zh0iAhNu5pSHM+S2nu1pXmkxpRNKcF76OOQMN3l9iG2lhQ6V+t3XoHnO Slb9zO/RLv6PcdInBB002A9hw9RGRg==
T8VEPKRQ5HEQDP39FSS3FN611CLMS6GM.cloud. 300 IN NSEC3 1 1 0 - 7936MI8L99IJ9020S7HUHM379F6OP13HJ NS DS RRSIG
T8VEPKRQ5HEQDP39FSS3FN611CLMS6GM.cloud. 300 IN RRSIG NSEC3 13 2 300 20260409091240 20260310123534 36804 cloud. llUD6TZrdzok1II7+pS8HgURtdnKUwkcBvcGyM2/G0tH2xfjuX5Vk6TG ByOtwureHD2V4c2nLAdqF2F4tLWNaA==
;; Received 581 bytes from 64.96.2.1#53(ns01.trs-dns.net) in 40 ms

;; UDP setup with 2600:9000:5304:d200::1#53(2600:9000:5304:d200::1) for vraj.pushkar.cloud failed: network unreachable.
;; UDP setup with 2600:9000:5301:e200::1#53(2600:9000:5301:e200::1) for vraj.pushkar.cloud failed: network unreachable.
vraj.pushkar.cloud.     300     IN      NS      ns-116.awsdns-14.com.
vraj.pushkar.cloud.     300     IN      NS      ns-1232.awsdns-26.org.
vraj.pushkar.cloud.     300     IN      NS      ns-1706.awsdns-21.co.uk.
vraj.pushkar.cloud.     300     IN      NS      ns-913.awsdns-50.net.
;; Received 187 bytes from 205.251.195.216#53(ns-984.awsdns-59.net) in 10 ms

;; UDP setup with 2600:9000:5306:aa00::1#53(2600:9000:5306:aa00::1) for vraj.pushkar.cloud failed: network unreachable.
```

---

4. Results and Observations

4.1 Self-Hosted DNS

- Responds correctly to queries
- Acts as authoritative server when NS points to EC2

4.2 Route 53 DNS

- Responds correctly
- Highly available and managed
- Becomes authoritative when NS updated

4.3 Key Observation

DNS resolution depends entirely on **NS records**

- When NS → EC2 → Self-hosted DNS active
- When NS → Route 53 → AWS DNS active

---

5. Comparison

Feature	Self-Hosted DNS	Route 53

Setup	Complex	Easy
Maintenance	Manual	Managed
Availability	EC2 dependent	High
Scalability	Limited	Automatic
Control	Full	Limited

---

6. Challenges Faced

- DNS propagation delay
- Incorrect NS configuration initially
- Local resolver confusion (127.0.0.53)
- Firewall (port 53) issues

---

7. Conclusion

This project successfully demonstrated:

- Implementation of both DNS systems
- DNS resolution behavior
- Importance of NS records

Switching NS records effectively changes the authoritative DNS server and controls how domain resolution works globally.

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8. Future Improvements

- Implement DNSSEC
- Add load balancing
- Use multiple DNS servers
- Automate deployment using Terraform

9. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____